Logarithms and Their Properties

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

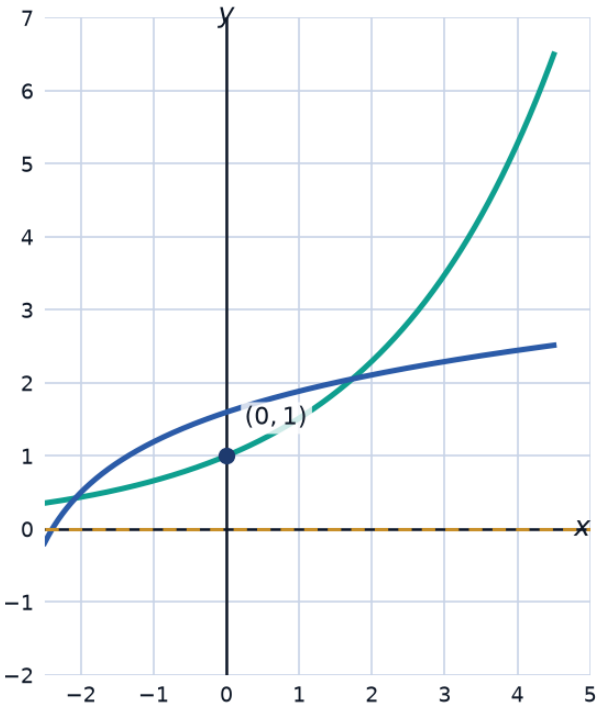
1 [Foundational] • Target: Evaluating logarithms

Evaluate each logarithm exactly:

a $\log_2 32$ **b** $\log_3 \frac{1}{9}$ **c** $\log 1000$ **d** $\ln e^7$

2 [Foundational] • Target: Evaluating logarithms

Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.



3 [Foundational] • Target: Evaluating logarithms

State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

4 [Foundational] • Target: Properties of logarithms

Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

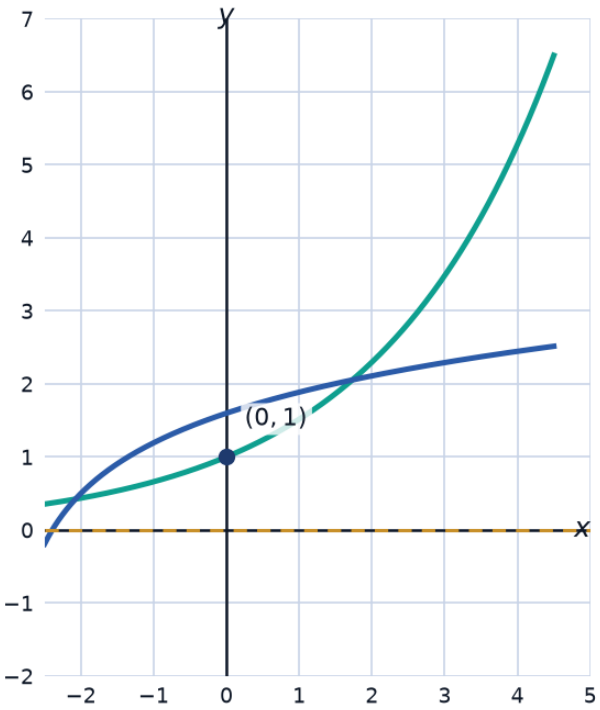
5 [Foundational] • Target: Properties of logarithms

Write as a single logarithm: $2\log x + 3\log y - \log z$.

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6 [Foundational] • Target: Properties of logarithms

Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:



a $\log_b 6$ **b** $\log_b 8$ **c** $\log_b 1.5$

7 [Foundational] • Target: Graphs of logarithmic functions

For $y = \log_2 (x + 4) - 1$, state:

a the domain **b** the vertical asymptote **c** the x-intercept

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphs of logarithmic functions

Describe how the graph of $g(x) = -\ln(x - 2) + 3$ is obtained from $y = \ln x$.

9 [Intermediate] • Target: Logarithmic scales and applications

The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

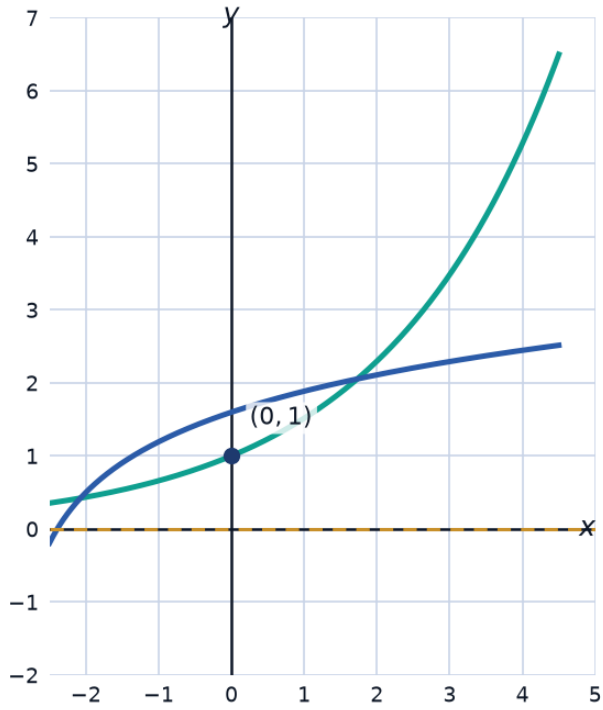
a An earthquake has $M = 6$. Express I in terms of I_0.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake?

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10 **[Intermediate]** • **Target: Logarithmic scales and applications**

Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?



11 **[Intermediate]** • **Target: More practice with logarithm properties**

Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

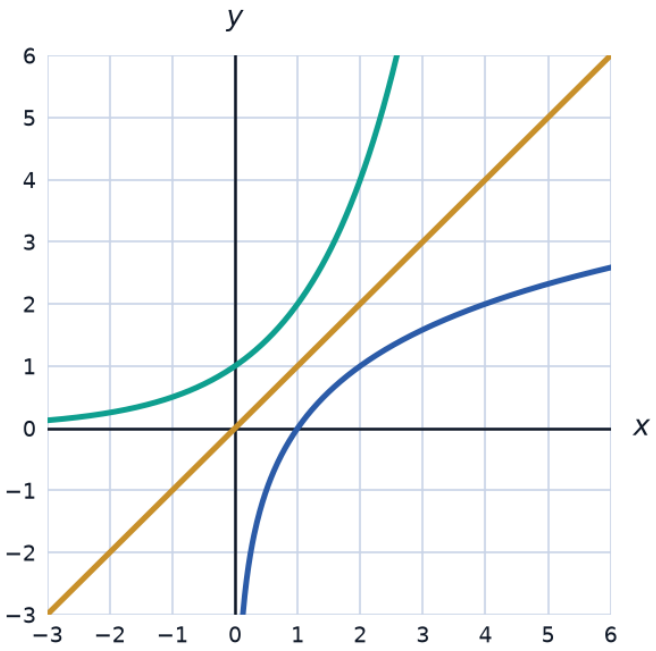
12 **[Intermediate]** • **Target: More practice with logarithm properties**

Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

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13 [Intermediate] • Target: Logarithms as inverse functions

The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.



14 [Intermediate] • Target: Solving simple logarithmic equations

Solve $\log_3 (x + 2) = 4$ for x.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

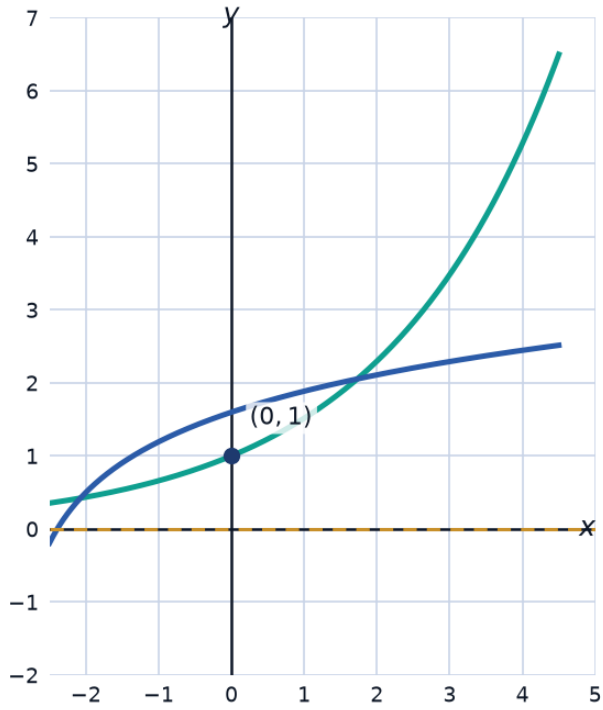
15 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $2\log x = \log(x + 6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.

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16 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 3.01 The Unit Circle and Exact Values

pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.



17 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.

18 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Explain why $\log(x + y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

19 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.

20 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 4.01 Circles and Parabolas

Find the projection of $\mathbf{u} = \langle 6, -2 \rangle$ onto $\mathbf{v} = \langle 2, -1 \rangle$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Definition of Logarithm and Inverses

Convert $\log_2(32) = 5$ to its equivalent exponential form.

a. $2^5 = 32$

b. $5^2 = 32$

c. $32^{1/5} = 2$

d. $2 \times 5 = 10$

22 [Foundational] • Target: Definition of Logarithm and Inverses

Evaluate $\log_3\left(\frac{1}{27}\right)$.

a. -3

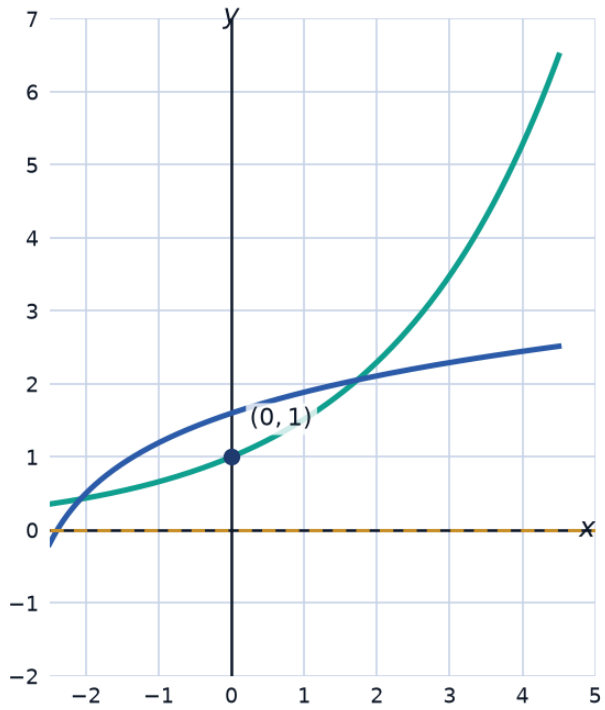
b. 3

c. $1/3$

d. $-1/3$

23 [Foundational] • Target: Definition of Logarithm and Inverses

What is the inverse function of $f(x) = e^{2x-1}$?



a. $f^{-1}(x) = \frac{\ln x + 1}{2}$

b. $f^{-1}(x) = \ln(2x - 1)$

c. $f^{-1}(x) = 2\ln x - 1$

d. $f^{-1}(x) = e^{-(2x-1)}$

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24 [Foundational] • Target: Product, Quotient, and Power Rules

Expand $\log_b\left(\frac{x^3y}{z^2}\right)$ into individual logarithms.

a. $3\log_b x + \log_b y - 2\log_b z$

b. $3\log_b x \cdot \log_b y - 2\log_b z$

c. $6(\log_b x + \log_b y - \log_b z)$

d. $\frac{3\log_b x + \log_b y}{2\log_b z}$

25 [Foundational] • Target: Product, Quotient, and Power Rules

Condense into a single logarithm: $2\ln(x) + \frac{1}{2}\ln(y) - 3\ln(z)$.

a. $\ln\left(\frac{x^2\sqrt{y}}{z^3}\right)$

b. $\ln(x^2 + \sqrt{y} - z^3)$

c. $\ln\left(\frac{2x \cdot \frac{1}{2}y}{3z}\right)$

d. $\ln(x^2y^{1/2}z^3)$

26 [Foundational] • Target: Product, Quotient, and Power Rules

Given $\log_a 2 \approx 0.301$ and $\log_a 3 \approx 0.477$, approximate $\log_a 18$.

a. 1.255

b. 0.778

c. 1.079

d. 0.954

27 [Foundational] • Target: Change of Base Formula

State the change of base formula to express $\log_b a$ in terms of natural logarithms.

a. $\log_b a = \frac{\ln a}{\ln b}$

b. $\log_b a = \frac{\ln b}{\ln a}$

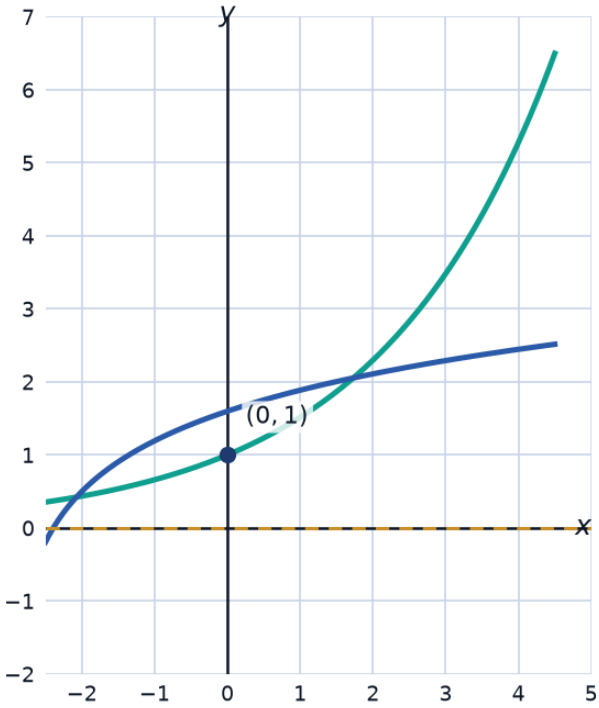
c. $\log_b a = \ln(a - b)$

d. $\log_b a = \ln(a) \cdot \ln(b)$

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28 [Foundational] • Target: Change of Base Formula

Evaluate $\log_4(32)$ using the change of base formula with base 2.



a. $5/2 = 2.5$

b. 8

c. 2

d. 5/4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Change of Base Formula

Evaluate the product $\log_2 3 \cdot \log_3 4 \cdot \log_4 8$.

a. 3

b. 2

c. 4

d. $\log_2 8$ is 3

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30 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Find the domain and vertical asymptote of $f(x) = \ln(2x - 6)$.

a. Domain: $(3, \infty)$, Asymptote: $x = 3$

b. Domain: $[3, \infty)$, Asymptote: $x = 3$

c. Domain: $(6, \infty)$, Asymptote: $x = 6$

d. Domain: $(0, \infty)$, Asymptote: $x = 0$

31 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

What is the range of any basic logarithmic function $f(x) = \log_b(x)$ $(b > 1)$?

a. $(-\infty, \infty)$

b. $(0, \infty)$

c. $[0, \infty)$

d. $(-\infty, 0)$

32 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Find the x-intercept of $y = \log_3(x + 2) - 1$.

a. $(1, 0)$

b. $(0, 0)$

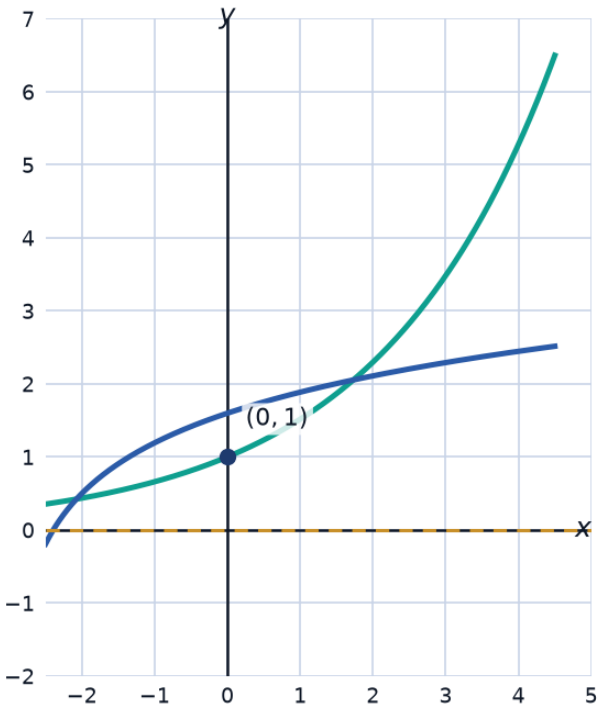
c. $(3, 0)$

d. $(-1, 0)$

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33 [Intermediate] • Target: Domains and Graphs of Logarithmic Functions

Expand $\log(x^3)$ using properties of logarithms.



a. $\log(x+y)$

b. $3\log x$

c. $\log(x)\cdot\log(y)$

d. $\frac{\log x}{\log y}$

34 [Intermediate] • Target: Evaluating logarithms

A 15 m rope is tied to the top of a pole and staked to level ground 9 m from the base of the pole. Find the angle the rope makes with the ground, to the nearest tenth of a degree.

a. 36.9°

b. 31.0°

c. 58.1°

d. 53.1°

35 [Intermediate] • Target: Properties of logarithms

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x)\cdot\log(y)$

c. $\log x+\log y$

d. $\log x-\log y$

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36 [Intermediate] • Target: Graphs of logarithmic functions

Expand $\ln\left(\frac{x^3}{y}\right)$ using properties of logarithms. (variant 4)

a. $3\ln x - \ln y$

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $3\ln x + \ln y$

37 [Intermediate] • Target: Logarithmic scales and applications

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 1)

a. $\frac{\log x}{\log y}$

b. $\log x - \log y$

c. $\log(x) \cdot \log(y)$

d. $\log x + \log y$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More practice with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Expand $\ln(xy^2)$ using properties of logarithms. (variant 3)

a. $\log(x) \cdot \log(y)$

b. $\ln x - 2\ln y$

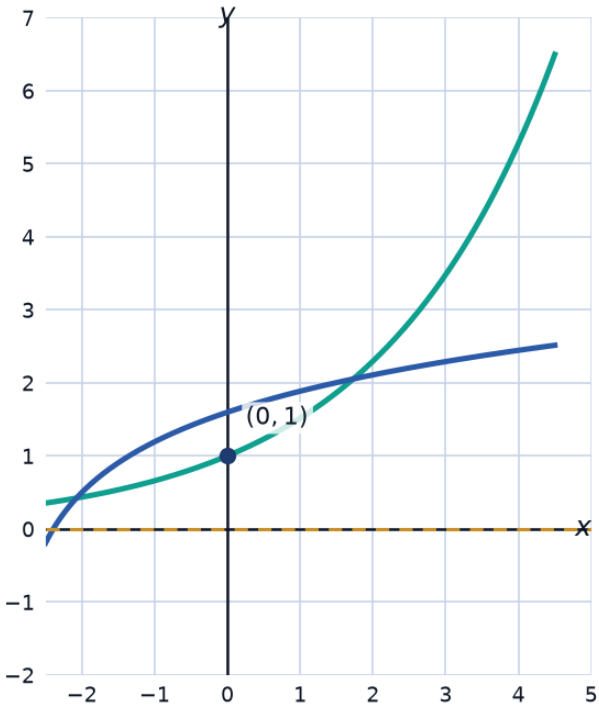
c. $\ln x + 2\ln y$

d. $\frac{\log x}{\log y}$

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39 [Challenging] • Target: Logarithms as inverse functions • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Expand $\ln(xy^2)$ using properties of logarithms. (variant 1)



a. $\ln x - 2\ln y$

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $\ln x + 2\ln y$

40 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 4.01 Circles and Parabolas

Expand $\log(xy)$ using properties of logarithms. (variant 6)

a. $\log(x) \cdot \log(y)$

b. $\log x + \log y$

c. $\log x - \log y$

d. $\frac{\log x}{\log y}$

41 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 4.02 Ellipses

Expand $\log(xy)$ using properties of logarithms. (variant 4)

a. $\log x - \log y$

b. $\frac{\log x}{\log y}$

c. $\log x + \log y$

d. $\log(x) \cdot \log(y)$

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42 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 4.03 Hyperbolas

Expand $\log(xy)$ using properties of logarithms.

a. $\log x - \log y$

b. $\log(x) \cdot \log(y)$

c. $\frac{\log x}{\log y}$

d. $\log x + \log y$

43 [Challenging] • Target: Definition of Logarithm and Inverses • Cross-Lesson: 5.01 Vectors: Components and Operations

Expand $\log(xy)$ using properties of logarithms. (variant 5)

a. $\log x - \log y$

b. $\frac{\log x}{\log y}$

c. $\log(x) \cdot \log(y)$

d. $\log x + \log y$

44 [Challenging] • Target: Product, Quotient, and Power Rules • Cross-Lesson: 5.02 The Dot Product

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

45 [Challenging] • Target: Change of Base Formula • Cross-Lesson: 5.03 Parametric Equations

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms. (variant 3)

a. $\frac{\log x}{\log y}$

b. $2\log x + 3\log y$

c. $\log(x) \cdot \log(y)$

d. $2\log x - 3\log y$

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